

Testing and Diagnostics

1. Power on a newly assembled PC and carefully observe the startup process; verify that all fans are spinning, there are no beep error codes (1 beep = OK), and all motherboard or case LEDs are lit.

-> PC Power-On and Startup Verification

1. Connect the PC to power

- Make sure the power cable is connected to the PSU.
- Connect the monitor and keyboard.
- Turn the PSU switch to **ON (I)**.

2. Power on the PC

- Press the case **Power** button.
- Observe the system carefully during startup.

3. Check the fans

- Verify that the **CPU cooler fan** and **case fans** start spinning.
- If the GPU has a zero-RPM mode, its fans may not spin continuously at idle; this can be normal depending on the GPU.

4. Check beep/error indicators

- Listen for motherboard beep codes if your system has a speaker/buzzer.
- **One short beep can indicate a successful POST on some systems**, but beep patterns vary by motherboard manufacturer.
- No beep does not necessarily mean there is a problem because many modern PCs do not have a motherboard speaker.

5. Check LEDs

- Observe the motherboard diagnostic LEDs, if present.

- Check case power/activity LEDs.
- Make sure there are no persistent **CPU, DRAM, VGA, or BOOT** diagnostic lights.

6. Check the display

- Confirm that the monitor displays the manufacturer's logo or BIOS/UEFI screen.
- If available, enter BIOS/UEFI and verify the CPU, RAM, and storage are detected.

Documentation Table

Check	Observation	Status
CPU fan	Spinning	☐ Pass
Case fans	Spinning	☐ Pass
GPU fans	Normal for the GPU model	☐ Pass
Beep/error code	Record actual beep pattern	☐ Pass
Motherboard LEDs	No persistent error LED	☐ Pass
Case LEDs	Power/activity LEDs functioning	☐ Pass
Display	BIOS/UEFI appears	☐ Pass

Suggested Report Paragraph

Startup Verification: I powered on the newly assembled PC and carefully observed the startup process. The CPU and case cooling fans were checked for normal operation, and the motherboard/case LEDs were inspected for abnormal error indicators. I also listened for POST beep codes and checked that the monitor displayed the startup/BIOS screen. The system was then checked in BIOS/UEFI to confirm that the major hardware components were detected correctly.

2. Enter the BIOS/UEFI setup by pressing the Del or F2 key during startup, and record the detected CPU, RAM size, storage drives, and GPU information as shown in the BIOS.

-> BIOS/UEFI Hardware Verification

1. **Start the PC** and immediately press **Del** or **F2** repeatedly during startup.

2. Once BIOS/UEFI opens, find the **System Information, Main, Hardware Information**, or similar section.
3. Record the hardware information displayed.

Hardware Record

Component	Information to Record
CPU	Model/name and, if shown, clock speed
RAM	Total installed capacity and speed
Storage Drive 1	SSD/HDD model and capacity
Storage Drive 2	SSD/HDD model and capacity, if present
GPU	Graphics card model, if displayed
BIOS/UEFI Version	Version number, if required

Example

CPU: Intel Core i5-12400

RAM: 16 GB DDR4

Storage: 1 TB NVMe SSD

GPU: NVIDIA GeForce GTX 1650

BIOS Version: [record the version shown]

 **Documentation:** Take a clear photo or screenshot of the BIOS/UEFI information screen showing the detected hardware.

Important: Replace the example specifications above with the **actual values shown on your PC**; BIOS menus and displayed information vary by motherboard manufacturer.

3.Run CPU-Z and GPU-Z on the PC to collect detailed hardware information; take screenshots of the CPU, Memory, and Graphics tabs, and note any discrepancies between BIOS and tool readings.

-> CPU-Z and GPU-Z Hardware Verification

1. **Run CPU-Z**

- Open **CPU-Z**.
- Take screenshots of:
 - **CPU tab** — processor model, cores/threads, clock speed.
 - **Memory tab** — RAM type, total size, channel mode, and DRAM frequency.
- 🖼️ **Screenshot 1: CPU-Z CPU tab**
- 🖼️ **Screenshot 2: CPU-Z Memory tab**

2. Run GPU-Z

- Open **GPU-Z**.
- Take a screenshot of the **Graphics Card** tab showing the GPU model, VRAM, bus interface, and other specifications.
- 🖼️ **Screenshot 3: GPU-Z Graphics Card tab**

Comparison With BIOS

Record the results in this table:

Hardware BIOS/UEFI Reading CPU-Z/GPU-Z Reading Discrepancy

CPU	_____	_____	None / _____
RAM	_____	_____	None / _____
Storage	_____	_____	None / _____
GPU	_____	_____	None / _____

Important Note About RAM Speed

CPU-Z's **DRAM Frequency** may appear approximately half the advertised DDR speed because DDR memory transfers data twice per clock cycle. For example, a reading around **1600 MHz** corresponds to an effective DDR4-3200 speed.

Suggested Report Paragraph

Hardware Verification: I used CPU-Z to record detailed information about the CPU and memory and GPU-Z to record graphics-card information. I compared these readings with the hardware information shown in BIOS/UEFI. The CPU, RAM, and GPU information was checked for

consistency, and any differences were recorded. Minor differences in displayed RAM frequency can occur because CPU-Z reports the actual memory clock while DDR memory's effective data rate is approximately twice that value.

4. Use Memtest86 or HWiNFO to perform a basic memory and system health diagnostic, then summarize the results and mention any warnings or errors reported.

Hint: If you see any errors, try reseating the RAM or checking BIOS settings, then rerun the test.

-> **Memory and System Health Diagnostic**

You can use **MemTest86** for a dedicated RAM test or **HWiNFO** for a broader hardware-health check.

Using MemTest86

1. Start **MemTest86** and begin the memory test.
2. Allow the test to complete at least one full pass.
3. Check the results for:
 - **Passes completed**
 - **Errors**
 - Memory detected
 - Test status
4. Take a screenshot of the final results.

Using HWiNFO

1. Open **HWiNFO** and select the hardware/system summary.
2. Check CPU, RAM, GPU, storage, temperatures, voltages, and fan readings.
3. Look for any warnings or abnormal readings.
4. Take a screenshot showing the diagnostic information.

Results Table

Test Item	Result
RAM detected	_____
MemTest86 passes	_____
Memory errors	0 / _____
CPU temperature	_____ °C
GPU temperature	_____ °C
Storage health	_____
Warnings	None / _____

If Errors Are Found

- Shut down the PC.
- Reseat the RAM module(s).
- Make sure the RAM is installed in the motherboard's recommended slots.
- Check BIOS memory settings and return them to appropriate/default settings if necessary.
- Run the diagnostic again.

Suggested Report Summary

Diagnostic Summary: I performed a basic memory and system-health diagnostic using MemTest86/HWiNFO. The system detected the installed RAM correctly, and the diagnostic was checked for memory errors and hardware warnings. The final results showed **[no errors/warnings OR describe the errors found]**. If an error was detected, the RAM was reseated and the relevant BIOS settings were checked before rerunning the diagnostic. The results after the second test were **[enter your actual result]**.

Use your actual test results in the brackets rather than assuming the test passed.

5. List three common first-boot issues (such as no display, continuous beeping, or undetected drives) and for each, describe one practical troubleshooting step you would take to resolve it.

-> **Common First-Boot Issues and Troubleshooting**

First-Boot Issue	Practical Troubleshooting Step
1. No display	Check that the monitor cable is connected to the correct video output. If a dedicated GPU is installed, make sure the monitor is connected to the GPU and that the GPU is fully seated in the PCIe slot.
2. Continuous beeping	Check the motherboard's manual for the specific beep-code meaning, then reseat the RAM and verify that it is fully locked into the recommended DIMM slot.
3. Storage drive not detected	Check both the SATA data cable and SATA power cable . Reconnect them firmly and then check the BIOS/UEFI storage section again.

Short Summary

During the first boot, common problems can include no display, continuous beep codes, and an undetected storage drive. For no display, I would check the monitor connection and GPU installation. For continuous beeping, I would identify the beep code using the motherboard manual and reseat the RAM. For an undetected drive, I would reconnect the SATA data and power cables and verify the drive in BIOS/UEFI.